Vehicle Service Data Contract

T1 | Star schema technical specification | Revision 1.0 | 3 October 2026

The vehicle service stores one tracked observation and its finalized plate, colour and manufacturer results in a shared analytical model. This contract defines the table meanings, identities, relationships and ingestion rules needed to implement that model consistently.

The supervisor-approved architecture is retained. This revision completes the technical rules; approval of these additions and database execution are not asserted.

Approved structure

Table group	Tables and purpose
Central fact	fact_vehicle_event anchors one observation in one camera and tracking run.
Recognition facts	fact_plate_recognition, fact_colour_recognition and fact_manufacturer_recognition store the three finalized outputs.
Shared dimensions	dim_camera, dim_date, dim_pipeline_run and dim_vehicle_type describe observation context.
Recognition dimensions	dim_plate_category, dim_colour, dim_manufacturer and dim_vehicle_model standardize result labels.

Row meanings and relationships

A vehicle event is one tracked lifetime within one camera and pipeline run. It is not a permanent identity for a physical vehicle. The same car on another camera or in a later processing run receives a separate event.

Each event has exactly one camera, date, pipeline run and vehicle type. Each recognition table contains at most one row per event, enforced by UNIQUE(vehicle_event_key). A completed ingestion writes all three result rows, including explicit unavailable or failed states. Each vehicle model belongs to one manufacturer.

What this revision resolves

Stable retry identity; one result per service and event; consistent pipeline-run keys; explicit result states; defined confidence provenance; UTC date assignment; manufacturer/model consistency; and the vehicle_type_key naming mismatch. Detailed frame history and individual OCR attempts remain outside the analytical tables.

Vehicle event identity and fields

Except for the calendar date_key, all *_key primary keys are BIGINT database-generated surrogate keys. UUID source identifiers are created before publication and reused unchanged on transport retries. Database keys are resolved by the writer and do not need to appear in producer messages.

fact_vehicle_event field	Type and rule
vehicle_event_key	BIGINT primary key.
observation_id	UUID NOT NULL UNIQUE. Stable identity of this tracked lifetime.
pipeline_run_key / camera_key	BIGINT NOT NULL foreign keys.
track_id / track_generation	BIGINT / INTEGER, NOT NULL. Generation starts at 1 and increases whenever a tracker ID is reused or reset within the same run and camera.
date_key	INTEGER NOT NULL foreign key. UTC date of first_seen.
vehicle_type_key	BIGINT NOT NULL foreign key; use the unknown member when unresolved.
first_frame / last_frame	BIGINT NOT NULL. Zero-based decoded-frame indices within this run and camera; last_frame >= first_frame >= 0.
first_seen / last_seen	TIMESTAMPTZ NOT NULL. UTC instants of the first and last frames that actually show the track.
sighting_time_basis	TEXT NOT NULL: host_frame_receipt or source_capture. One basis per event; source_capture requires trusted capture timestamps.
track_duration_seconds	DOUBLE PRECISION NOT NULL, finite and >= 0. Elapsed sighting interval measured with a monotonic clock; not time spent in recognition.
frames_observed	INTEGER NOT NULL, >= 1 and <= last_frame - first_frame + 1. Count contributing frames, not the full frame span.
final_reason	TEXT NOT NULL: track_lost, end_of_input, service_stop, camera_outage or tracker_reset.
finalized_at	TIMESTAMPTZ NOT NULL. Host UTC time when the snapshot was closed.

Required uniqueness

UNIQUE(pipeline_run_key, camera_key, track_id, track_generation) prevents a second source identity for the same tracked lifetime. UNIQUE(vehicle_event_key, pipeline_run_key) supports composite foreign keys from the recognition facts. A conflicting identity or payload is an error, not an update.

first_seen <= last_seen is required. Clock discontinuities must be detected before publication. Sighting timestamps are not compared with host finalized_at when their clocks differ. An abrupt crash may leave no final event; T3 must define durable publication and recovery behavior.

Plate and colour recognition fields

Each recognition fact has a BIGINT primary key, a NOT NULL vehicle_event_key, and a NOT NULL pipeline_run_key. Its pair (vehicle_event_key, pipeline_run_key) references the same pair in fact_vehicle_event. This version stores outputs from the event run only; a rerun creates a new run and new event identities.

fact_plate_recognition field	Type and meaning
plate_fact_key	BIGINT primary key.
plate_text	TEXT nullable. Normalized complete plate text; leading zeros retained. Never an integer.
recognition_state / error_code	TEXT state for plate-number recognition; nullable bounded error code, never an exception dump.
plate_category_key	BIGINT NOT NULL FK to dim_plate_category; unknown member when unresolved.
plate_category_state / plate_category_error_code	TEXT category state and nullable bounded error code, independent of plate-number recognition.
confidence / confidence_metric	Nullable score and metric for plate_text, not for plate category.
category_confidence / category_confidence_metric	Nullable category score and metric; not derived from the OCR score.
selected_crop_frame	BIGINT nullable. Frame index in the event interval; null when no crop was selected.
selected_crop_score / selected_crop_score_metric	Nullable finite quality score and its named scale; not automatically a probability.
recognition_attempts	INTEGER nullable, >= 0. Number of plate-crop OCR attempts, including failed attempts; 0 means none, null means unmeasured.
local_latency_ms	Nullable finite nonnegative wall duration of OCR on the selected crop; includes preprocessing and decoding, excludes queue wait.
total_recognition_latency_ms	Nullable finite nonnegative elapsed time from first submitted plate attempt to result resolution, including queueing and retries.

fact_colour_recognition field	Type and meaning
colour_fact_key	BIGINT primary key.
colour_key	BIGINT NOT NULL FK to dim_colour; unknown member when unresolved.
colour_state / error_code	TEXT state and nullable bounded error code.
colour_confidence / colour_confidence_metric	Nullable score and its meaning for the stored colour label.
colour_samples	INTEGER nullable, >= 0. Number of frame-level colour outputs included in aggregation.
type_confidence / type_confidence_metric	Nullable score for the exact vehicle-type label referenced by the event. A latest-frame score for another label must not populate it.

The existing colour location for type_confidence is retained for compatibility with the approved design. It does not change the vehicle type stored on the event.

Manufacturer results and pipeline runs

fact_manufacturer_recognition field	Type and meaning
manufacturer_fact_key	BIGINT primary key.
vehicle_event_key / pipeline_run_key	NOT NULL foreign keys with the same composite run-consistency rule as the other recognition facts.
manufacturer_key	BIGINT NOT NULL FK to dim_manufacturer; unknown member when unresolved.
manufacturer_state / manufacturer_error_code	TEXT result state and nullable bounded error code.
model_key	Nullable BIGINT FK to dim_vehicle_model. Null unless a model is known.
model_state / model_error_code	TEXT state independent of manufacturer inference, and nullable bounded error code.
manufacturer_confidence / manufacturer_confidence_metric	Nullable score and named metric for the selected manufacturer.
model_confidence / model_confidence_metric	Nullable score and named metric for the selected vehicle model.
observations_count	INTEGER nullable, >= 0. Number of frame/crop outputs included in manufacturer aggregation.

When model_key is present, manufacturer_state and model_state must both be known. The pair (model_key, manufacturer_key) must reference dim_vehicle_model(model_key, manufacturer_key), backed by a matching UNIQUE constraint. This prevents a model being attached to the wrong manufacturer. If manufacturer is known and model recognition is unavailable, keep the manufacturer and set model_state = unsupported.

Pipeline execution context

dim_pipeline_run field	Type and meaning
pipeline_run_key / run_id	BIGINT primary key / UUID NOT NULL UNIQUE. One coordinated processing execution; all three outputs use the same run.
service_name / input_source	Nonempty TEXT. Stable service and source reference; omit embedded credentials.
output_video	Nullable TEXT artifact reference. Annotated video is optional.
frames_processed	Nullable BIGINT >= 0; decoded frames actually processed across the run.
duration_seconds	Nullable finite DOUBLE PRECISION >= 0; monotonic elapsed run duration.
processing_fps	Nullable finite DOUBLE PRECISION >= 0; frames_processed / duration_seconds when duration is positive.
metrics	JSONB object, default {}. Record code commit, model artifact versions or hashes, configuration fingerprint, start/end UTC times and diagnostic metrics.

A run row can be created before processing and completed when it ends. Unmeasured or unfinished run totals remain null. Model/configuration provenance is recorded before output publication and is not silently replaced. These are run-level measures: do not sum them after joining to vehicle events.

Dimensions and controlled values

Dimension labels use stable canonical values. Each recognition state conveys why a result is unavailable; the unknown dimension member alone never carries that explanation.

Dimension	Fields and constraints
dim_camera	camera_key PK; camera_id TEXT NOT NULL UNIQUE; camera_name TEXT; location TEXT; stream_type TEXT; active BOOLEAN NOT NULL. camera_id is stable and contains no connection secret.
dim_date	date_key INTEGER PK in YYYYMMDD form; full_date DATE UNIQUE NOT NULL; day, month, month_name, quarter, year, weekday. weekday uses ISO numbering, Monday = 1.
dim_vehicle_type	vehicle_type_key BIGINT PK; vehicle_type_name TEXT NOT NULL UNIQUE. This corrects the diagram label dim_vehicle_type used as a column name.
dim_plate_category	plate_category_key BIGINT PK; category_name TEXT NOT NULL UNIQUE. Detailed Tunisian/foreign category vocabulary.
dim_colour	colour_key BIGINT PK; colour_name TEXT NOT NULL UNIQUE.
dim_manufacturer	manufacturer_key BIGINT PK; manufacturer_name TEXT NOT NULL UNIQUE.
dim_vehicle_model	model_key BIGINT PK; manufacturer_key BIGINT NOT NULL FK; model_name TEXT NOT NULL. UNIQUE(manufacturer_key, model_name) and UNIQUE(model_key, manufacturer_key).

Vocabulary rules

Vehicle type, colour, plate category and manufacturer each have a reserved unknown member. A known result must reference a non-unknown member. Unknown keys are resolved by canonical label, not assumed numeric constants. Model uses a null key with an explicit state when no model is known.

The original category and vehicle-type lists are examples, not a verified exhaustive taxonomy. Store the selected vocabularies and source-label mappings in a versioned seed manifest before freeze. Do not split grouped source colours such as silver/gray into a specific colour without supporting evidence.

The current binary Tounes word-marker result is not a detailed plate category. Neither marker absence nor its other label may be mapped automatically to foreign, private or another plate category. Keep detailed category unsupported until an actual classifier or validated rule supplies it.

Metadata changes

Version 1 camera metadata reflects current descriptive values. Historical location changes are not preserved by this simple dimension. Category and manufacturer meanings must not be reassigned; label aliases map to canonical members. Human corrections and training labels remain outside prediction fact rows.

Recognition states and score semantics

State	Meaning and required values
known	A supported value was resolved. Plate text is nonempty; categorical key is non-unknown; model_key is present for a known model.
inferred_unknown	Inference completed but did not resolve an accepted value. Result value is null or the unknown member; confidence is null.
not_attempted	Capability exists but inference did not run, including no usable input. Result value is null or the unknown member; confidence is null.
unsupported	The producer has no capability for this attribute. Result value is null or the unknown member; confidence is null.
inference_error	An attempted step failed without a usable final result. Result value is null or the unknown member; confidence is null; error code is required.

Every finalized event publishes all recognition states, including separate plate-category and model states. Earlier failed attempts do not force inference_error when a later attempt produced a usable final result. Error codes are null outside inference_error; attempt diagnostics may retain transient errors.

Confidence rules

All confidence fields use DOUBLE PRECISION. Supplied values must be finite and in [0, 1], with a nonempty paired *_metric name. Both fields are null when no defensible score is available. A known result may legitimately have null confidence.

Metric names describe how scores were produced: for example ocr_recognition_score or temporal_winner_weight_share. Neither is automatically calibrated probability or measured accuracy. Different metrics are not averaged together.

The current latest-frame vehicle detector score must not be attached to a different locked vehicle-type label. Store type_confidence as null unless its provenance supports that exact label. A selected crop quality score has its own named scale and is not subject to the confidence range unless that metric defines it.

Plate text and time conventions

Plate normalization is a versioned function shared by the producer and evaluator. It preserves leading zeros and meaningful symbols, and documents handling of whitespace and category markers. Raw OCR text may remain in diagnostic evidence; it must not silently replace normalized plate_text. Version 1 requires its exact normalization rules to be recorded before freeze.

Timestamp fields represent UTC instants. Observation date is derived from first_seen in UTC, even if the event ends on another day. Camera-local calendar reports must convert explicitly. first_seen and last_seen describe visible observations; finalized_at describes processing closure. Current process-local monotonic timestamps cannot be relabelled as UTC.

Event transport and database ingestion

Finalized event envelope

Publish one complete finalized message per observation with schema_version, message_id UUID, observation_id UUID, run_id UUID, camera_id, track_id, track_generation, event_at UTC, and the event plus all recognition results. Categorical values use canonical labels; the database writer resolves their keys. Retries preserve all identifiers and semantic content.

message_id identifies the transport message; observation_id identifies the tracked lifetime. Version 1 has one finalized message per observation. Active updates may serve a live interface but do not create analytical facts. Store this envelope schema with the table specification; no operational event-history table is required by this analytical model.

Transaction rules

1. Validate the version, field types, vocabulary mappings, states, scores, source identity and run consistency before writing.

2. Resolve camera, date, run and controlled dimension keys. Serialize ingestion for observation_id, including its first delivery. Do not create arbitrary dimension labels from unvalidated predictions.

3. If observation_id already exists, compare all immutable event and recognition fields. An identical retry is a no-op; changed content is a conflict. Also reject the same run/camera/track/generation tuple presented with a different observation_id.

4. Insert the event and its three recognition rows in one transaction. The unique event and per-service constraints prevent duplicate analytical rows. Roll back the entire transaction on an error.

5. Commit before acknowledging the transport message. A crash after commit and before acknowledgment is handled by the identical-retry rule. Conflicts follow an explicit operational error path.

Recognition facts are immutable after successful ingestion in version 1. Reprocessing uses new run and observation identities; analytics must select the intended run when comparing results or counting vehicles. Database run summaries may be finalized separately. Detailed recovery, transport durability and monitoring are implemented in T3.

Database and writer responsibilities

The database enforces primary/foreign keys, source-identity uniqueness, one recognition row per event, matching run keys, manufacturer/model consistency, nonnegative counts and basic state/score constraints. The writer enforces complete messages, canonical normalization, trusted time provenance and atomic creation of all recognition rows. Direct fact writes must be restricted to the guarded ingestion path.

Completion checks

Before freezing the contract, record acceptance of the added rules and the versioned vocabulary/normalization manifests. Before deployment, execute the migration and verify identical replay, conflicting replay, tracker-ID reuse, transaction rollback, missing capability states, mismatched run/model keys and UTC date boundaries. This report does not claim those database tests have run.